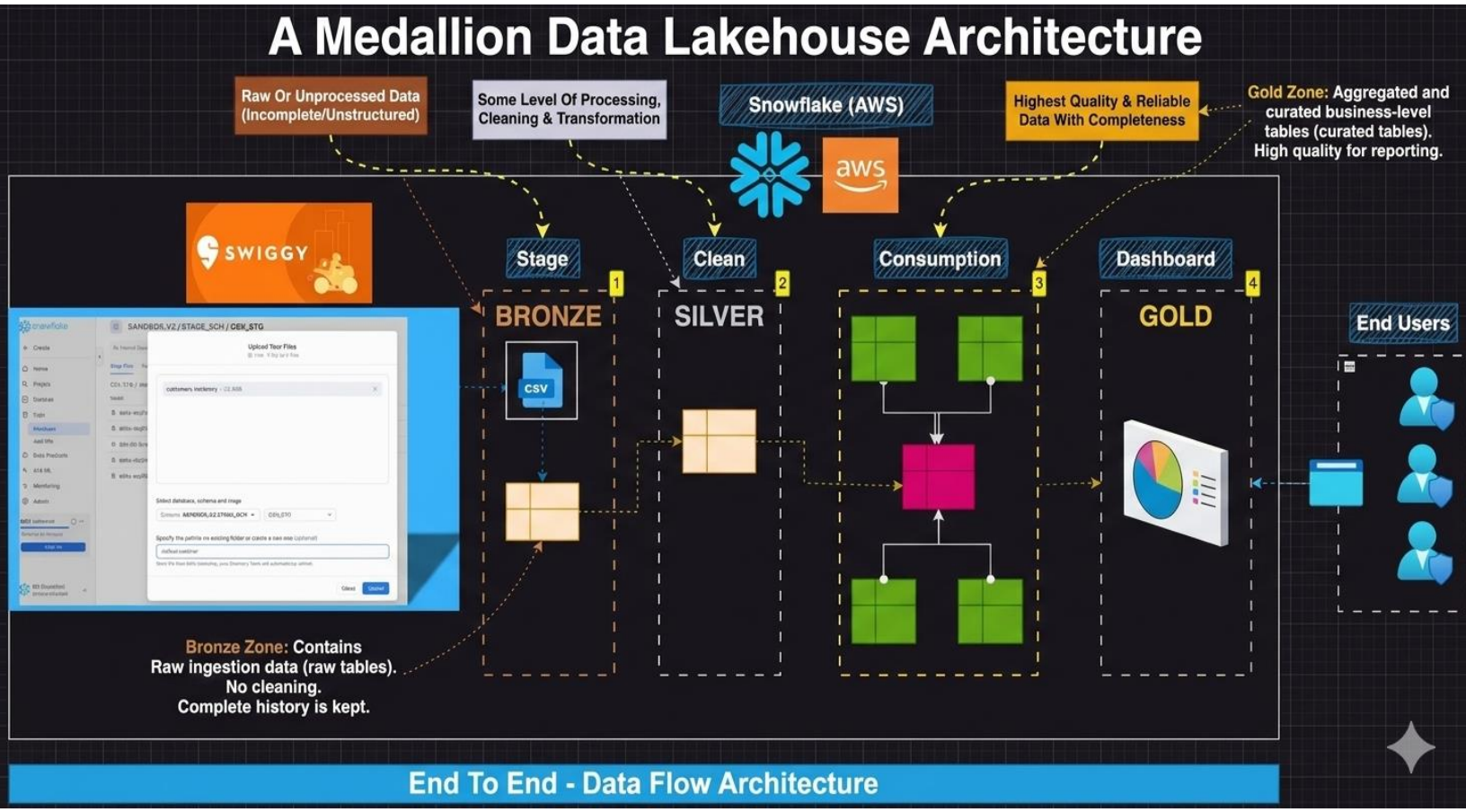
A Medallion Data Lakehouse Architecture



STAR & SNOWFLAKE SCHEMA HYBRID

Snowflake Schema Highlight: Tables like **'location'** and **'delivery'** are normalized with their own dimension-to-dimension relationships. **'orders'** connects to **'delivery'**, which connects to **'location'**. The 'Dim. Context' label points to this deeper normalization, defining this as a Snowflake structure because dimensions are hierarchically organized.

Star Schema Highlight: The relationship between **'orders'** and **'customer'** is direct and denormalized. One central fact table (**'orders'**) connects directly to its dimensions, defining this as a **Star** structure because it has a simple, star-like hub-and-spoke direct dimension layout.

Dim. Context

Primary Fact Table: orders
- Records transaction metrics (e.g., `totalamount`).

Dimension Table: customer
- Context for the orderer.

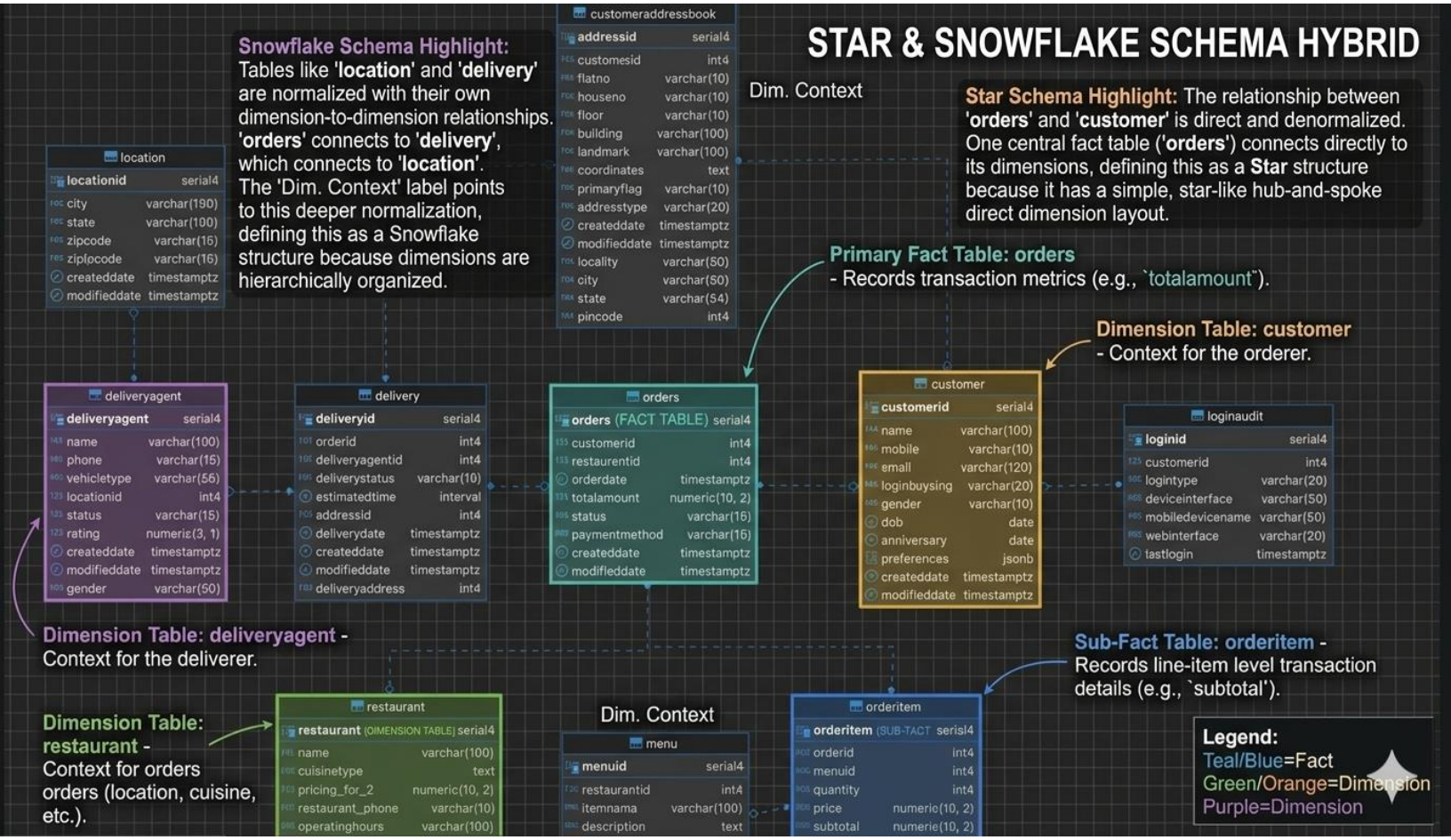
Dimension Table: deliveryagent - Context for the deliverer.

Sub-Fact Table: orderitem - Records line-item level transaction details (e.g., `subtotal`).

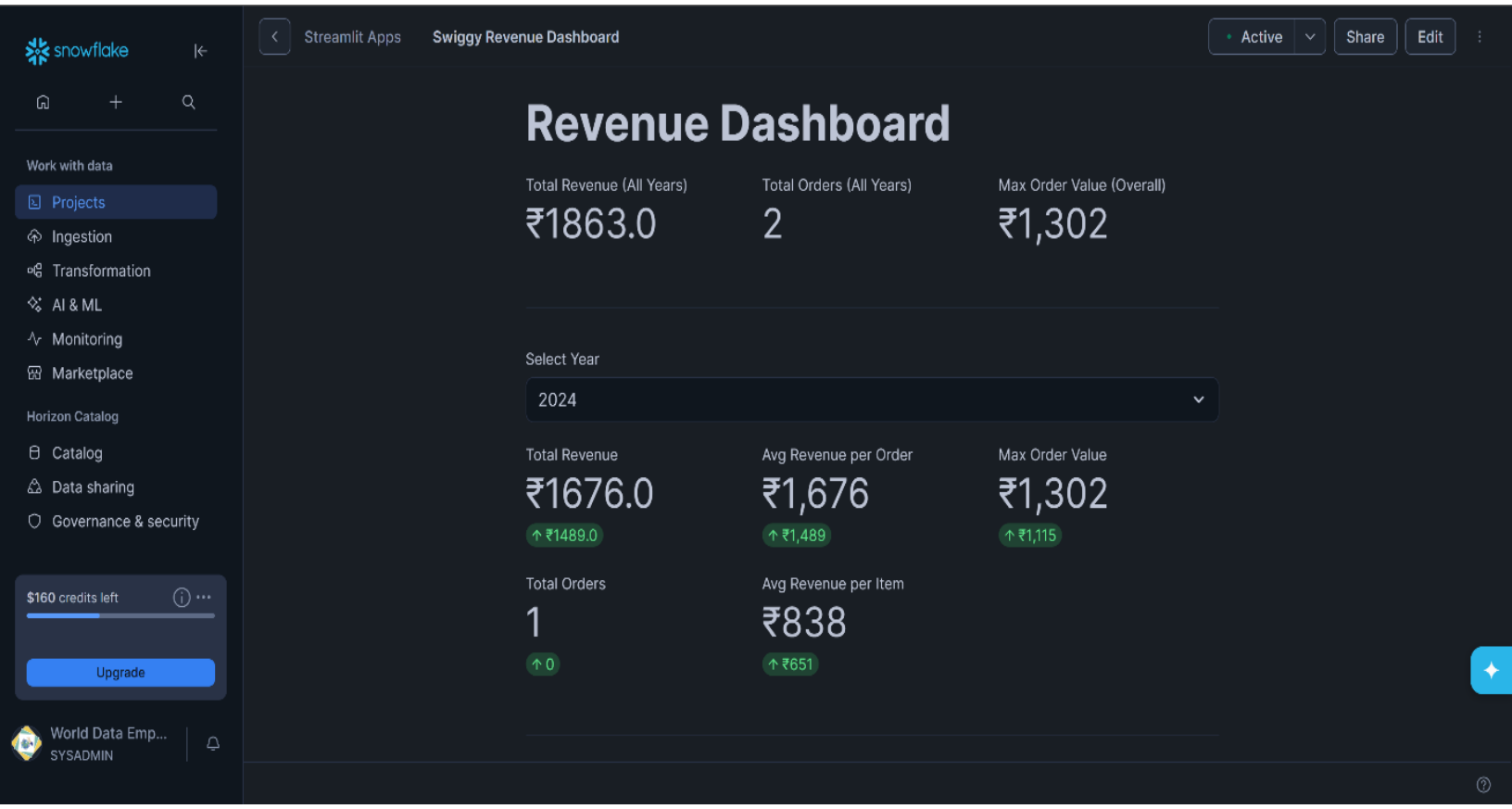
Dimension Table: restaurant - Context for orders orders (location, cuisine, etc.).

Dim. Context

Legend:
Teal/Blue=Fact
Green/Orange=Dimension
Purple=Dimension



RESTAURANT
Restaurant Entity
MENU
Menu Entity
User
Login Entity
DATA ENGINEERING SIMPLIFIED
Location Entity
SWIGGY
FOOD DELIVERY APP
TRISTATE
Customer Entity
Delivery Entity
Address Entity
Delivery Agent Entity
Order Entity


snowflake
Streamlit Apps
Swiggy Revenue Dashboard
Active
Share
Edit
Work with data
Projects
Ingestion
Transformation
AI & ML
Monitoring
Marketplace
Horizon Catalog
Catalog
Data sharing
Governance & security
$160 credits left
Upgrade
World Data Emp...
SYSADMIN
Revenue Dashboard
Total Revenue (All Years)
₹1863.0
Total Orders (All Years)
2
Max Order Value (Overall)
₹1,302
Select Year
2024
Total Revenue
₹1676.0
↑ ₹1489.0
Avg Revenue per Order
₹1,676
↑ ₹1,489
Max Order Value
₹1,302
↑ ₹1,115
Total Orders
1
↑ 0
Avg Revenue per Item
₹838
↑ ₹651